

The Balanced Readout

Why the winner-take-all territory diverges from the constrained mass, and the dual-offset correction that removes the divergence

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Abstract

Phase 1 minimises a Γ -convergence energy over continuous densities and enforces equal area as a constraint on the *continuous masses*. The partition it delivers, however, is the *winner-take-all territory* of those densities. This document derives the exact identity relating the two, shows why nothing in the variational problem controls their difference, and analyses the two artifacts that follow — territory far from the equal-area target, and territory that splits into disconnected pieces. It then derives the correction implemented in `src/partition/balanced_readout.py`: a per-cell additive offset obtained from the dual of a transportation problem. Three properties are proved: the corrected readout grows each cell *monotonically outward from its own core* (the locality that in-flow corrections lacked); the attainable balance is bounded below by the one-vertex granularity, and this bound is a theorem rather than an observation, because the underlying linear program is *not* integral; and the offsets control area but not connectivity, which is why a second, connectivity-preserving pass is required. Where an argument is a model rather than a derivation — notably the worst-cell scaling in N , and the mechanism behind disconnection — this is stated explicitly.

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1 Overview

The method of Bogosel and Oudet [2] represents a partition of a surface into N cells by N density functions and minimises a Modica–Mortola-type functional [8, 7] whose Γ -limit is the total interface length. Equal areas are imposed as constraints on the integrals of those densities.

To hand a partition downstream one must convert the densities into a definite assignment of each mesh vertex to one cell. The natural choice, and the one used throughout this project, is pointwise *winner-take-all*: each vertex joins whichever cell has the largest density there. We refer to this conversion step as the *readout*, by analogy with instrumentation, where the internal state of a device and the value it reports are distinct objects and the conversion between them is a separate source of error.

The central observation of this document is that the constrained quantity and the delivered quantity are *different functionals of the same field*, and that their difference is exactly computable, is supported entirely on the diffuse interface band, and is referenced nowhere in the energy or the constraints. It is therefore free to drift, and it drifts further as the number of cells grows at fixed mesh resolution.

Remark 1.1. This is a statement about the readout at finite ε and finite mesh. The Γ -convergence result concerns the limit of the *energy* and is not in question. Nothing here contradicts [2]; the readout is a practical choice whose error term had not been quantified.

Related documents: docs/math/06-phase1-energy-discretization/ derives the discrete energy itself; docs/math/07-phase1-wta-balance/ derives a *rejected* attempt to enforce balance inside the flow, and its Proposition 1 is the identity restated as (5) below; docs/reference/winner.take.all.partition.gap.m is the standing prose account.

2 Notation and Setup

Let the surface be triangulated with vertex set $\{1, \dots, V\}$ and let M be the P1 finite-element mass matrix. Write

$$v_i = \sum_j M_{ij} \tag{1}$$

for the lumped mass of vertex i , so that $v_i > 0$ and $\sum_i v_i = |\Sigma|$, the total surface area. The densities are $u_{ik} \in [0, 1]$ for vertex i and cell $k \in \{1, \dots, N\}$, subject to the partition-of-unity and equal-area constraints

$$\sum_{k=1}^N u_{ik} = 1 \quad \forall i, \quad \sum_{i=1}^V v_i u_{ik} = \bar{A} = \frac{|\Sigma|}{N} \quad \forall k. \tag{2}$$

Definition 2.1 (Readout and territory). The *readout* is the map $\omega : \{1, \dots, V\} \rightarrow \{1, \dots, N\}$,

$$\omega(i) = \arg \max_k u_{ik}, \tag{3}$$

ties broken by lowest index. The *territory* of cell k is $\Omega_k = \{i : \omega(i) = k\}$ and its *discrete area* is

$$T_k = \sum_{i \in \Omega_k} v_i. \tag{4}$$

The constraint in (2) fixes $\sum_i v_i u_{ik}$; validity is graded on T_k . These coincide only if every u_{ik} is exactly 0 or 1.

3 The readout gap is exact

Proposition 3.1 (Territory–mass identity). *For every cell k , with the constraint (2) satisfied,*

$$T_k - \bar{A} = \underbrace{\sum_{i \in \Omega_k} v_i (1 - u_{ik})}_{\text{gain}_k} - \underbrace{\sum_{i \notin \Omega_k} v_i u_{ik}}_{\text{lost}_k}. \quad (5)$$

Proof. Split the constrained sum over the territory and its complement:

$$\bar{A} = \sum_i v_i u_{ik} = \sum_{i \in \Omega_k} v_i u_{ik} + \sum_{i \notin \Omega_k} v_i u_{ik} = \left(T_k - \sum_{i \in \Omega_k} v_i (1 - u_{ik}) \right) + \text{lost}_k,$$

using $\sum_{i \in \Omega_k} v_i u_{ik} = T_k - \sum_{i \in \Omega_k} v_i (1 - u_{ik})$ from (4). Rearranging gives (5). $\square$

The identity is exact: no linearisation, no assumption on ε or on the mesh. Its two terms have a direct reading. gain_k is mass that cell k *owns but does not hold* — vertices it wins while its density there is below one. lost_k is mass it *holds but does not own* — density deposited at vertices won by somebody else.

Remark 3.1 (The gap lives entirely in the interface band). Both terms vanish wherever the field is crisp. Deep inside cell k , $u_{ik} \approx 1$, so $1 - u_{ik} \approx 0$ and the vertex contributes nothing to gain_k . Far outside, $u_{ik} \approx 0$ and it contributes nothing to lost_k . Only vertices in the diffuse band, where $0 < u_{ik} < 1$, contribute to either. *The readout gap is a band phenomenon*, and everything about how it scales follows from how much of a cell the band occupies.

3.1 Why nothing controls it

The energy is a function of u ; the constraints (2) are linear in u . Neither references ω . Moreover the map $u \mapsto \omega$ is piecewise constant and therefore discontinuous: it is constant on each cell of the arrangement $\{u_{ik} - u_{il} = 0\}$ and jumps across it. Consequently

- T_k has zero gradient with respect to u almost everywhere, so no first-order optimality condition can constrain it;
- adding T_k to the objective is not possible without a surrogate, and the surrogate used in `docs/math/07-phase1-wta-balance/` was measured and rejected (see §6.3);
- the quantity $\text{gain}_k - \text{lost}_k$ is, from the optimiser’s point of view, an unobserved by-product.

4 Artifact I: territory far from the equal-area target

4.1 The band fraction

Let $R_{\text{cell}} = \sqrt{|\Sigma|/(\pi N)}$ be the radius of a disc of the target area and let h be a representative edge length. In the discretisation used here the interface width parameter is tied to the mesh, $\varepsilon \sim h$ (`docs/math/06-phase1-energy-discretization/`). A cell’s band is a collar of width $O(\varepsilon)$ around a boundary of length $O(R_{\text{cell}})$, so the fraction of a cell occupied by band is

$$f_b \sim \frac{\varepsilon \cdot 2\pi R_{\text{cell}}}{\pi R_{\text{cell}}^2} = \frac{2\varepsilon}{R_{\text{cell}}} \sim \frac{h}{R_{\text{cell}}}. \quad (6)$$

With $h \sim \sqrt{|\Sigma|/V}$ this becomes

$$f_b \sim \sqrt{\frac{N}{V}} = \left(\frac{V}{N}\right)^{-1/2}, \quad (7)$$

i.e. **the band fraction is governed by vertices per cell alone**. This recovers, from geometry, the empirical form $f_b = 3.72/\sqrt{2V/N}$ recorded in `docs/reference/winner_take_all_partition_gap.md`.

Remark 4.1 (The scaling trap). At *fixed mesh*, raising N shrinks R_{cell} and therefore *raises* f_b : the band becomes a larger share of each cell precisely as the cells become the objects one cares about. Holding f_b fixed requires $V \propto N$ — constant vertices per cell — so the mesh must grow linearly in the number of cells merely to keep the readout error from worsening.

4.2 The diffuse runt

Equation (5) permits a configuration that is feasible, low-energy, and useless: a cell whose density is spread as a broad halo, everywhere slightly below its neighbours'. It then satisfies $\sum_i v_i u_{ik} = \bar{A}$ exactly while winning almost nothing, because lost_k is large and gain_k small. Such a cell is *confidently in the right place and insufficiently loud*. Empirically at $N = 300$ the worst cell reached 36.15% below target while its continuous mass was exact.

4.3 Worst-of- N — a model, not a derivation

Remark 4.2 (Explicitly not established). The following is the argument by which the project has explained why the problem sharpens with N . It is a **model**. The distribution of $\text{gain}_k - \text{lost}_k$ across cells has never been measured, and the available data points differ in both λ and mesh, so they cannot test it.

If one models each cell's net exchange as a random variable with mean zero and standard deviation $\sigma \propto f_b$, roughly independent across cells, then validity — which is graded on $\max_k |T_k - \bar{A}|$ — is an extreme-value statistic, and for light-tailed exchange the maximum of N draws grows like $\sigma\sqrt{2\ln N}$. Two consequences follow *if the model holds*: the worst cell degrades slowly but without bound as N grows even at fixed f_b ; and any mechanism that merely *reduces the variance* of the exchange postpones the problem rather than removing it. Only a mechanism that makes the delivered representation balanced *by construction* is immune to the tail. That prediction is consistent with every measured outcome in this project, which is evidence for the model but not a proof of it.

5 Artifact II: disconnected territory

Remark 5.1 (Weaker than §4, and deliberately so). The existence of this artifact is a theorem (§5.1); its *cause* in our runs is a hypothesis that has never been traced to a specific event. The two are separated below.

5.1 Nothing forbids it

The readout (3) is defined pointwise. Ω_k is a super-level set of the function $i \mapsto u_{ik} - \max_{l \neq k} u_{il}$, and a super-level set of an arbitrary function on a graph need not be connected. Neither the energy nor the constraints contain any term that is finite for connected territories and infinite otherwise. Disconnection is therefore *admissible* and costs nothing directly; a split cell is a legitimate local minimiser of the relaxed problem.

What makes it pathological is external: a minimal-perimeter cell is connected, so a split territory is a non-physical artifact of the representation, and Phase 2 cannot consume it, since a split cell yields multi-loop contours.

Remark 5.2 (Why the other two gates are blind to it). A split cell's pieces sum to the correct area and each piece is crisp. It therefore passes a dormant-cell check (peak density ≈ 1) and an area-imbalance check (total area correct). Connectivity is a third, independent validity dimension.

5.2 The mechanism — hypothesis

The constraint $\sum_i v_i u_{ik} = \bar{A}$ is enforced by an iterative projection onto the feasible set. The projection is a *nonlocal* operator: correcting a deficit in cell k adds a rank-one term spread across the whole free set, followed by renormalisation. Raising u_{ik} everywhere by a near-uniform amount can therefore flip the argmax at vertices arbitrarily far from the cell’s core — wherever cell k happened to be a close second — and open a second basin.

Three observations are consistent with this and none establishes it:

1. Splits appear *while balance is improving*. At $N = 300$ they are born mid-level-2, at the same time as the imbalanced count falls from 234 to 10: balance is being bought with disconnection.
2. Mechanisms that push balance through the projection make it dramatically worse. A discrete-area trim produced 14/200 split cells against a matched control’s 0/200; a soft area constraint produced 3–4/100.
3. The correction derived in §6, which is *local* by Proposition 6.2, does not manufacture splits.

Remark 5.3 (What would settle it). Decompose the per-iteration update into its gradient and projection contributions at the iteration a split is born, and test whether the new component’s vertices had u_{ik} raised by the projection. This has not been done; it is recorded as future work in docs/plans/PUBLICATION_READINESS_PLAN.md.

6 The corrected readout

6.1 The assignment problem

Fix any per-vertex score matrix s_{ik} ; the shipped readout uses $s_{ik} = \log u_{ik}$, with densities clipped from below so that exact zeros give a finite, very negative score. We seek a hard assignment maximising total score subject to every cell attaining its target area:

$$\max_x \sum_{i,k} v_i s_{ik} x_{ik} \quad \text{s.t.} \quad \sum_k x_{ik} = 1 \quad \forall i, \quad \sum_i v_i x_{ik} = \bar{A} \quad \forall k, \quad x_{ik} \geq 0. \quad (8)$$

This is a transportation problem: supplies v_i at vertices, demands $\bar{A}$ at cells, and both marginals sum to $|\Sigma|$, so it is feasible.

Remark 6.1 (Terminology). This project’s prose has called (8) a *semi-discrete* optimal transport problem. Strictly it is *discrete-to-discrete*: V weighted points to N cells. The semi-discrete problem — a continuous source measure transported to N atoms, in the sense of [1, 6, 5] — is its continuum limit as the mesh refines. The offsets derived below are the discrete analogue of the power weights in that literature.

6.2 The dual, and where the offsets come from

Introduce multipliers φ_i for the row constraints and ψ_k for the column constraints. Stationarity of the Lagrangian in x_{ik} gives $v_i s_{ik} - \varphi_i - v_i \psi_k \leq 0$, with equality wherever $x_{ik} > 0$, so $\varphi_i = v_i \max_k (s_{ik} - \psi_k)$ and the dual is

$$\min_{\psi \in \mathbb{R}^N} D(\psi), \quad D(\psi) = \sum_i v_i \max_k (s_{ik} - \psi_k) + \bar{A} \sum_k \psi_k. \quad (9)$$

D is a sum of pointwise maxima of affine functions plus a linear term, hence **convex** and piecewise linear. Complementary slackness forces every vertex onto a maximising cell, so an optimal assignment is

$$\omega_\psi(i) = \arg \max_k (s_{ik} - \psi_k). \quad (10)$$

Remark 6.2 (Sign convention). The implementation stores $\psi^{\text{code}} = -\psi$ and evaluates $\arg \max_k (s_{ik} + \psi_k^{\text{code}})$. All statements below use the code convention, in which *raising* ψ_k *grows* cell k .

Proposition 6.1 (The implemented update is the dual subgradient). *At any ψ where the argmax is unique at every vertex, D is differentiable with*

$$\frac{\partial D}{\partial \psi_k} = \bar{A} - T_k(\psi), \quad T_k(\psi) = \sum_{i: \omega_\psi(i)=k} v_i, \quad (11)$$

and in general this is a subgradient. Hence gradient descent on D is exactly “grow the cells that are too small, shrink the cells that are too large”.

Proof. Differentiating the first term of (9) at a point of uniqueness picks out, for each i , the maximising index $\omega_\psi(i)$, contributing $-v_i$ to the ψ_k derivative precisely when $\omega_\psi(i) = k$; the second term contributes $\bar{A}$. Summing gives (11). At points where the argmax is non-unique, D is a maximum of finitely many affine functions and the same expression is an element of the subdifferential for any measurable tie-breaking. $\square$

Code: dual ascent on the offsets

```
src/partition/balanced_readout.py: solve_dual_offsets(scores, lumped_mass, target,
config)
psi -= (eta0 / (1 + decay*it)) * rel, with rel = (areas - target)/target — the nor-
malised form of (11). The routine returns its best iterate, since  $D$  is piecewise linear and subgradient
descent is not monotone. Scores enter only through argmax(scores + psi), so the routine is
correct for any additive score matrix; only its step schedule is scale-dependent.
```

6.3 Locality: cells grow outward from their own cores

Proposition 6.2 (Nested monotone growth). *Fix ψ_{-k} and regard $\Omega_k(\psi_k)$ as a function of the single offset ψ_k . Define the margin of vertex i against cell k as*

$$m_i = \max_{l \neq k} (s_{il} + \psi_l) - s_{ik}. \quad (12)$$

Then $\Omega_k(\psi_k) = \{i : m_i \leq \psi_k\}$. Consequently $\psi_k < \psi'_k$ implies $\Omega_k(\psi_k) \subseteq \Omega_k(\psi'_k)$, and vertices are acquired in nondecreasing order of m_i .

Proof. Immediate from (10): $i \in \Omega_k$ iff $s_{ik} + \psi_k \geq \max_{l \neq k} (s_{il} + \psi_l)$, i.e. iff $\psi_k \geq m_i$. The sets are sublevel sets of the fixed function m , hence nested. $\square$

This is the property the in-flow corrections lacked, and it is worth stating plainly what it does and does not say.

- **What it gives.** Growth is *ordered by margin*. A cell claims first the vertices where it was most nearly winning, which — for a density field that is unimodal about the cell’s core — are the vertices adjacent to the territory it already holds. Correction proceeds outward through the adjacent band rather than being deposited wherever the algebra puts it.
- **What it does not give.** Proposition 6.2 is a statement about the *order* of acquisition, not about connectivity. If the margin function m has two separated basins — exactly the situation Artifact II describes — then nested growth will populate both. Locality prevents the correction from *manufacturing* new components; it cannot repair components that the density field already possesses. Hence §7.

Contrast this with enforcing balance inside the flow. There the correction reaches the field through the constraint projection, which is nonlocal (§5); measured outcomes were a net regression in both attempts (docs/experiments/04-territory-aware-highn-validation/, docs/experiments/05-soft-area-constra

6.4 The granularity floor is a theorem, not an observation

It is tempting to assert that (8) has an integral optimum because transportation problems do. That reasoning is unavailable here.

Proposition 6.3 (Non-integrality). *The classical integrality theorem for the transportation polytope requires integer supplies and demands. In (8) the supplies are the real lumped masses v_i and the demands are $\bar{A} = |\Sigma|/N$. Unless some subset $S \subseteq \{1, \dots, V\}$ satisfies $\sum_{i \in S} v_i = \bar{A}$ exactly, no integral feasible point exists at all, and every vertex of the polytope splits at least one vertex’s mass between cells. A basic feasible solution of a transportation problem with V sources and N sinks has at most $V + N - 1$ nonzeros, so at most $N - 1$ vertices are split.*

Consequently the exact optimum of (8) is in general *fractional*, and the readout (10) is a rounding of it. The residue is bounded by the mass of the split vertices, which yields the floor

$$\min_{\psi} \max_k \frac{|T_k(\psi) - \bar{A}|}{\bar{A}} \geq (\text{generically}) \frac{\max_i v_i}{\bar{A}} \approx \frac{N}{V} \cdot \frac{\max_i v_i}{\bar{v}}, \quad (13)$$

where $\bar{v} = |\Sigma|/V$. **A vertex is indivisible, so no assignment method whatever can do better.** This is why acceptance thresholds in this project are stated as multiples of $\max_i v_i / \bar{A}$ rather than as absolute percentages: a bar below (13) is not demanding, it is impossible.

Remark 6.3 (Measured values). $\max_i v_i / \bar{A}$ is 1.011% at $V = 47,488$, $N = 300$; 0.421% at $V = 114,144$, $N = 300$; 0.140% at $V = 114,144$, $N = 100$; and 0.561% at $V = 114,144$, $N = 400$. The floor scales as N/V — inversely with vertices per cell — for the same reason (7) does.

7 Connectivity repair

The offsets control area but, by the second bullet of §6.3, not connectivity. A second pass restores it. It operates on labels only.

Stage 1 — island absorption. For each cell, compute the connected components of its territory in the mesh edge graph. Every non-largest component whose area exceeds a fixed fraction of $\bar{A}$ is relabelled to the neighbouring cell with which it shares the longest boundary. This step is unconditionally safe: a component adjacent to cell c can always be merged into c without disconnecting c , since the merged set is connected through the shared boundary. On exit every cell is connected (or empty, which is reported).

Stage 2 — area restoration by boundary transfer. Absorption perturbs the areas, so equal areas are restored by moving *single boundary vertices* from richer cells to poorer ones. A move of vertex i from donor d to receiver r is admitted only if

$$T_d - T_r > v_i, \quad (14)$$

which guarantees that the transfer strictly reduces $|T_d - \bar{A}| + |T_r - \bar{A}|$ rather than overshooting, and only if $\Omega_d \setminus \{i\}$ remains connected — an exact articulation test, cheap because it is local to one cell. Candidates are ranked by the score margin $s_{ir} - s_{id}$, so the least costly transfers are taken first.

Remark 7.1 (What is and is not guaranteed). Both invariants — every cell connected, and no admissible improving move remaining — hold on exit *by construction*. What is **not** guaranteed is that the exit state attains the granularity floor (13): stage 2 is a greedy local search over single-vertex moves, and a configuration can be locally optimal without being globally balanced, particularly when (14) blocks every candidate. The implementation therefore reports its own strain — number of moves, moves blocked by the connectivity test, sweeps used, and whether the sweep cap was reached — so that a run beyond what repair can fix says so rather than returning a degraded partition silently.

Code: the two-stage readout

```
src/partition/balanced_readout.py:    apply_balanced_readout(...)    orchestrates
solve_dual_offsets → reassign_islands → rebalance_boundaries; _stays_connected
is the articulation test and build_vertex_adjacency the edge graph, built from faces so that it
respects the true surface topology (e.g. the torus periodic wrap) rather than a flat parametrisation.
CLI: scripts/balanced_readout.py.
```

8 Relation to power diagrams

With scores $s_{ik} = -d^2(i, z_k)$ for sites z_k , (10) reads $\omega(i) = \arg \max_k (-d^2(i, z_k) + \psi_k) = \arg \min_k (d^2(i, z_k) - \psi_k)$, which is precisely a *power diagram* (Laguerre tessellation) with weights ψ_k . That the weights can be chosen to prescribe the cell capacities is Aurenhammer, Hoffmann and Aronov’s Minkowski-type theorem [1]; modern solvers for the continuous version are given by Mérigot [6] and Kitagawa, Mérigot and Thibert [5], and the construction is used for blue-noise sampling by de Goes *et al.* [4].

The balanced readout is the same algebra applied to a different score. Two differences are worth recording:

1. Our scores are $\log u$, a *relaxation output*, not a distance. There are no sites, and no geometric structure is assumed — only that the score is additive in ψ .
2. Our problem is finite-to-finite, so the damped Newton methods of [5], which exploit the smoothness of the semi-discrete objective in the continuous-source setting, do not transfer directly; D in (9) is piecewise linear, hence nonsmooth. Subgradient descent is used instead. Entropic regularisation [3] is an alternative that was implemented and measured for this problem, and lost to subgradient descent on five of six score matrices (docs/experiments/07-phase0-shared-harness/).

Remark 8.1 (We are not aware of this being reported). Applying a capacity-constrained assignment as a *readout correction for a Γ -convergence relaxation* — as opposed to as a construction in its own right — is not something we have found in the literature. Stated as a population claim, the corpus searched is this project’s bibliography plus two full-text readings; that is not a systematic review, and the claim is phrased accordingly.

9 Measured effect

Recorded here for completeness; the study that measures them belongs in docs/experiments/, which does not yet contain one for this stage.

torus, $N = 300$, $V = 47,488$	source	after readout	
imbalanced cells (gate 5%)	10	0	
worst cell deviation	36.15%	1.63%	floor 1.011%
disconnected cells	2	0	
label-boundary length	189.73	193.30	+1.88%
wall time	—	≈ 17 s	

The +1.88% boundary cost is the price of the correction: territory is moved to meet the area target, and moved territory has a longer border. Phase 2 subsequently recovers most of it.

Summary table

Quantity	Status	Where	Implementation
Territory–mass identity (5)	derived, exact	§3	—
Band fraction (7)	derived, scaling	§4	—
Worst-of- N growth	model, unmeasured	§4	—
Disconnection admissible	derived	§5.1	—
Disconnection <i>mechanism</i>	hypothesis, untraced	§5	—
Dual (9), subgradient (11)	derived	§6	<code>solve_dual_offsets</code>
Nested growth (locality)	derived	§6.3	—
Granularity floor (13)	derived from non-integrality	§6	<code>vertex_granularity</code>
Island absorption	derived, safe	§7	<code>reassign_islands</code>
Boundary transfer	greedy; invariants only	§7	<code>rebalance_boundaries</code>

References

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